Thesis Prospectus

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What Hurts Retrieval-Augmented Generation in Biomedical Question Answering: A Stress-Test Study of Noise, Conflict and Unanswerability

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Abstract

Retrieval-Augmented Generation (RAG) is gaining traction in biomedical Question Answering (QA) for generating responses from scientific literature; however, these systems remain prone to significant errors when the retrieved information is flawed. RAG systems may lead to dangerous outcomes in medicine by drawing incorrect conclusions from scientific evidence or by misinterpreting the patient population. Despite this risk, current QA systems are primarily evaluated with aggregate metrics (e.g., accuracy) that do not identify the evidentiary faults behind harmful behaviors such as overconfidence or unsupported claims. To fill this gap in the literature on the current state of biomedical QA systems, a diagnostic stress test will be performed on best-in-class biomedical RAG pipelines using expert-reviewed benchmarking datasets from BioASQ. Reliability will be evaluated by systematically introducing three controlled evidence-based "stressors": noise from retrieval, conflicting sources of information (e.g., studies that contradict each other), and the impossibility of finding the desired information due to inadequate evidence to conclude. The primary purpose of this study is not to introduce new QA models, but to take a diagnostic approach and use quantitative and qualitative measures beyond surface-level similarity between model-generated responses and truthful answers in BioASQ. Another goal is to evaluate how well the generated responses align with the existing evidence and to assess the likelihood of mixing conflicting sources or making unsupported guesses. By isolating these evidentiary weaknesses, this study identifies which evidence conditions most strongly drive unsafe behaviors and provides practical recommendations for safer biomedical RAG.

Keywords: Biomedical QA, RAG, AI, Groundedness, Stress Testing

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CHAPTER 1

INTRODUCTION

1.1 Background

Biomedical question answering (QA) supports clinicians, researchers, and students by providing evidence-based answers and differs from general QA because mistakes, such as a wrong conclusion, the wrong patient population, or outdated information, can have serious consequences (Malley et al., 2011). Since biomedical knowledge evolves rapidly, purely parametric models not connected to external data can struggle to stay current.

Retrieval-Augmented Generation (RAG) has therefore become common in biomedical QA. A RAG system retrieves relevant scientific text (for example, PubMed abstracts) and generates an answer based on that evidence. In theory, this should improve both factuality and traceability, because the answer can be checked against supporting documents rather than relying solely on what the model “remembers” (Si et al., 2024). In practice, however, biomedical RAG can still fail in ways that are not captured well by a single accuracy score or ROUGE (Abo El-Enen et al., 2025). Systems may produce confident, fluent answers that are poorly supported or unsupported when the retrieved context is imperfect.

Three major sources of misinformation that frequently lead to system failure are: (1) noise generated from searches by using poorly correlated or misleading snippets; (2) conflicting results from retrieved sources that point to different conclusions; and (3) missing answers from retrieved context because of a lack of supporting evidence. The above scenarios are possible in large biomedical text mines and can have a significant impact on evidence-based QA. The resulting responses may be consistent and confident, but they may only reflect the erroneous information used to create them, leading to unnecessary confusion, misinterpretation, and an inability to generate reliable responses due to insufficient evidence.

This study adopts a diagnostic approach rather than proposing a novel retrieval model. It uses the BioASQ (Anastasia et al., 2025) benchmark to stress-test a baseline evidence-based QA pipeline under controlled versions of these three conditions. It assesses reliability metrics, such as how much of the generated answer is truly supported by retrieved evidence and what kinds of systematic errors appear under each condition, in addition to similarity to reference answers. The objective is to determine which evidentiary problems most severely "hurt" biomedical RAG and to provide practical guidance for building safer, more trustworthy biomedical QA systems.

CHAPTER 2

LITERATURE REVIEW

2.1 Biomedical QA benchmarks and the role of BioASQ

The BioASQ initiative was introduced as a large-scale challenge for biomedical semantic indexing and QA, explicitly targeting systems that can answer natural-language biomedical questions using evidence from scientific articles and structured resources (Tsatsaronis et al., 2015). Biomedical question answering (QA) differs from open-domain QA because questions often reflect expert information needs and answers should be grounded in biomedical literature.

A major asset for research is the BioASQ-QA dataset, which provides English biomedical questions along with "gold" (reference) answers and related material, and was designed to reflect realistic expert needs and be more challenging than many general QA benchmarks (Krithara et al., 2023).

These properties make BioASQ an appropriate evaluation environment for studying reliability and failure modes of evidence-grounded biomedical QA systems.

2.2 Retrieval-Augmented Generation as the baseline paradigm

Retrieval-Augmented Generation (RAG) formalizes a general approach in which a model retrieves relevant documents from an external corpus and generates an answer conditioned on the retrieved evidence. RAG was proposed to improve performance on knowledge-intensive tasks and to support provenance, while also recognizing that factuality and reliability depend strongly on retrieval quality and evidence use (Lewis et al., 2020).

Because the objective of this thesis is to identify "what hurts RAG" in biomedical QA, RAG provides a natural baseline that links document retrieval, evidence selection, and final answer generation.

2.3 Biomedical retrieval and domain-specific language models

Biomedical text is characterized by specialized terminology, synonymy, and frequent abbreviations. Domain-specific pretraining has been shown to improve a wide range of biomedical NLP tasks, supporting the use of biomedical encoders in downstream pipelines (Jin et al., 2023).

Beyond general biomedical language models, biomedical information retrieval (IR) has also seen specialized retriever–reranker architectures. MedCPT is a representative example of a contrastively trained biomedical retriever and reranker built using large-scale PubMed search logs, and it reports strong performance across multiple biomedical IR tasks (Jin et al., 2023).

These works motivate using biomedical-aware retrieval components when building baseline systems for diagnostic studies, since retrieval quality is a major determinant of evidence-grounded QA outcomes.

2.4 Evidence stressors that "hurt" RAG in biomedical QA

2.4.1 Noise and evidence contamination

A common failure mode for RAG-style systems is **noisy retrieval**, where retrieved documents are only weakly related or contain distracting information. In retrieve-then-generate pipelines, irrelevant or partially relevant contexts can cause models to produce fluent but incorrect answers, particularly when the generator over-trusts retrieved text. The RAG framework itself highlights provenance and factuality as open challenges

In biomedical QA, this issue can be amplified by lexical overlap between conditions, drugs, and outcomes, creating "near-miss" evidence that appears relevant but does not support the required claim.

2.4.2 Conflict and contradictory contexts

A distinct stressor occurs when retrieved contexts disagree, which is realistic in large corpora and especially plausible in scientific literature. QACC ("Question Answering with Conflicting Contexts") demonstrates that a substantial fraction of unambiguous questions can yield conflicting evidence when retrieving from large collections, and benchmarks show that strong LLMs have limitations in resolving conflicts reliably (Liu et al., 2024).

While QACC is not biomedical-specific, it provides methodological grounding for treating conflict as an independent evaluation condition, rather than folding it into generic retrieval noise. This is important because conflict can lead to a particularly harmful failure mode: answers that confidently assert one side or implicitly merge contradictory claims.

2.4.3 Insufficient evidence and unanswerable questions (abstention)

Another major failure mode is **insufficient evidence**, where the provided context does not contain enough information to answer the question. SQuAD 2.0 formalizes this by introducing unanswerable questions and emphasizing that QA systems should learn to recognize when no supported answer exists, rather than guessing (Rajpurkar et al., 2018).

This "know what you don't know" framing directly motivates biomedical RAG evaluation under missing-evidence conditions, where abstention or uncertainty handling is necessary to prevent hallucinated medical claims.

2.4.4 Measuring groundedness and reliability beyond ROUGE/F1

ALCE proposes an evaluation setting for LLM-generated answers with citations and introduces automatic metrics for citation quality, encouraging reproducible assessment of whether answers are properly grounded (Gao et al., 2023).

FActScore proposes a fine-grained factuality evaluation method by decomposing generations into atomic facts and computing the proportion supported by a trusted source, motivating supported-claim

style metrics for long-form answers (Min et al., 2023).

RAGAS provides a reference-free framework for evaluating RAG pipelines along dimensions such as faithfulness and relevance, which is useful when gold references are incomplete or when analyzing many stress-test variants (Es et al., 2024)

Finally, CoVe (Chain-of-Verification) represents a mitigation direction: models can draft an answer, generate verification questions, answer them independently, and revise the response, reducing hallucinations across tasks (Dhuliawala et al., 2024)

Together, these works enable a thesis evaluation that goes beyond surface-level similarity and explicitly measures whether generated biomedical answers remain supported when evidence is noisy, conflicting or missing.

2.5 Research Gap

1. Current evaluations are mostly based on overall performance metrics (such as accuracy, F1, ROUGE) and enhancements at the component level, but do not separate the effects of various evidence conditions on system behavior. Consequently, it is still uncertain if the main factors behind failures in biomedical RAG are retrieval noise, conflicting evidence, or a lack of adequate supporting evidence.
2. While noisy retrieval and missing evidence have received some attention, **conflicting evidence**—a common and realistic scenario in biomedical literature—remains understudied in controlled evaluation settings. The ways in which RAG systems handle contradictory sources, such as selectively choosing a side or blending incompatible claims, are not well characterized.
3. Frameworks for diagnostic evaluation in biomedical retrieval-augmented generation (RAG) that go beyond mere answer similarity and actively evaluate aspects such as evidence grounding, support, and systematic error patterns (such as unsupported claims, blending of conflicting information, or guessing when uncertain) using expert-curated benchmarks like BioASQ are currently insufficient.

This dissertation tackles these shortcomings by conducting a controlled stress test of a biomedical RAG pipeline in the presence of noise, conflict, and unanswerability. In this assessment, both conventional metrics and evidence-based reliability analyses are utilized.

2.6 Research Question

1. Which evidence-based stressor—retrieval noise, conflicting evidence, or unanswerable questions has the most significant negative impact on the reliability of biomedical RAG systems?
2. How well do standard automatic metrics (accuracy, F1, ROUGE) track evidence-based reliability outcomes under controlled stressors, and where do they systematically diverge?

CHAPTER 3

METHODOLOGY

3.1 Overview

Biomedical question-answer retrieval systems are currently governed by Retrieval Augmented Generation (RAG) as a leading paradigm since such systems allow for answers that rely on existing scientific literature rather than a knowledge base based only on model parameters (Abo El-Enen et al., 2025). However, recent studies have shown that RAG systems can still produce errors such as unsupported claims, selection bias, and synthesis errors under situations where the retrieval has noisy, conflicting, or missing evidence (Zhou et al., 2024). The majority of current assessments on biomedical QA systems are calculated using an aggregate set of tasks metrics (e.g., accuracy, F1, ROUGE) and therefore do not provide insight into the underlying evidence conditions that contribute to harmful behaviors by the models.

To overcome this limitation, a framework for controlled stress testing of biomedical RAG systems. We are not modifying the retrieval component or the generation component in isolation; instead, we will fix both the retriever and the generator and systematically vary the evidence provided to the generator. This allows us to conduct a causal analysis of the impact of different evidence conditions (including noise, conflict, and lack of support) on the correctness, grounding, and reliability of the generated answers.

An overview of our proposed methodologies is shown in Figure 3.1. The proposed methodology consists of five stages: S1: create a controlled biomedical corpus; S2: use dense retrieval plus reranking; S3: manipulate the evidence for each answer; S4: use constraints when generating answers; and S5: conduct a multi-faceted evaluation of the resulting answers.

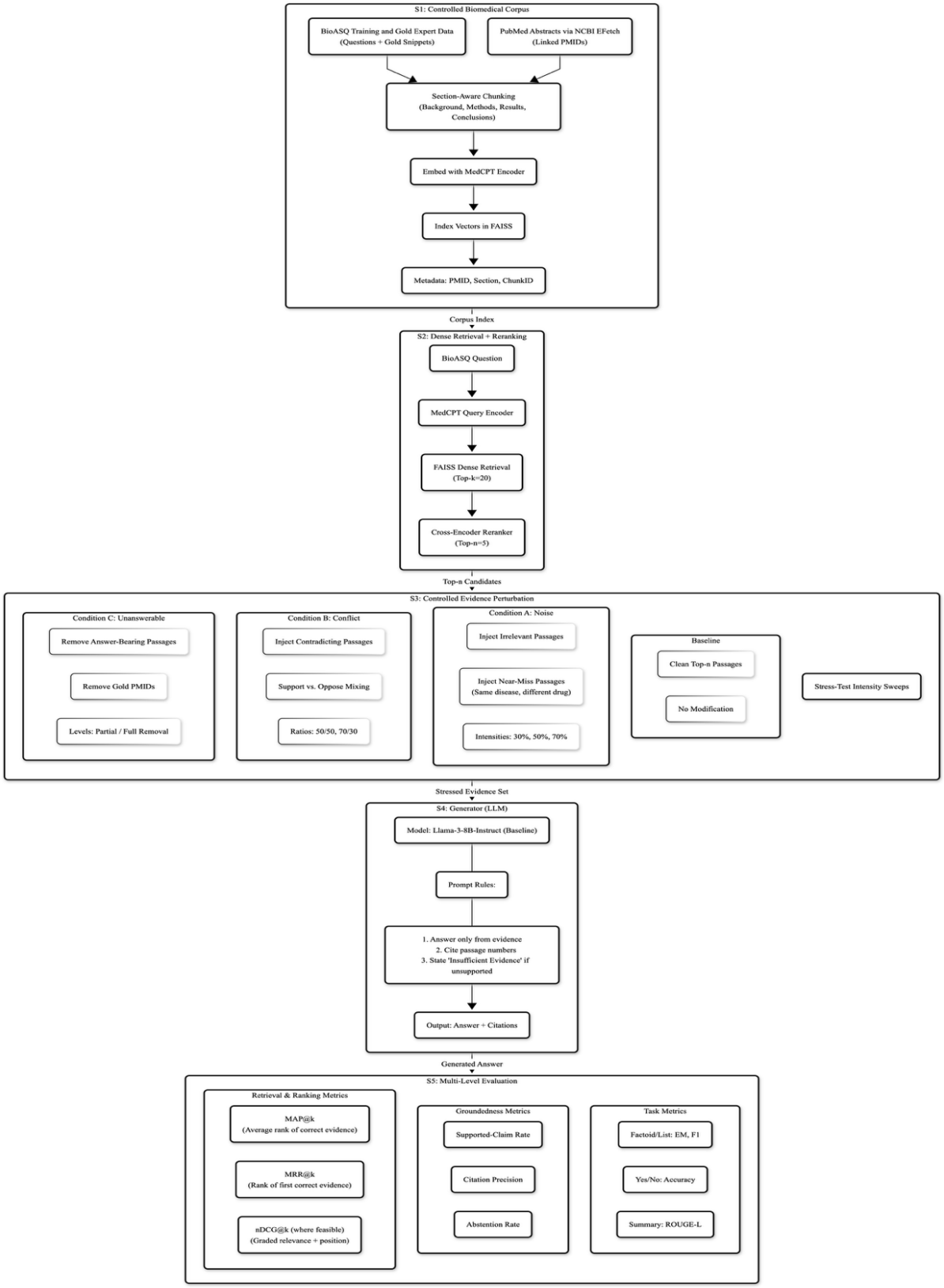

Fig 3.1: End-to-End Architecture for Retrieval-Augmented Biomedical Question Answering with Multi-Level Evaluation.

3.2 Step 1: Controlled Biomedical Corpus Construction

Create a controlled biomedical corpus by utilizing questions and related materials from the BioASQ training and golden datasets, which serve as a widely recognized benchmark for biomedical semantic question answering. In the initial closed-corpus configuration, the corpus is generated directly from evidence provided by BioASQ, which includes expert-linked PubMed identifiers (PMIDs) and text snippets annotated by experts. Later, enhance the corpus by fetching relevant PubMed abstracts for BioASQ-linked PMIDs using the NCBI PubMed EFetch API in a subsequent extension. This allows for evaluation on a larger evidence pool while maintaining alignment with the BioASQ reference sources.

All documents are segmented using section-aware chunking, which divides content into structural areas like background, methods, results, and conclusions. Each chunk is treated as a separate evidence unit and associated with metadata, including PMID, section label, and chunk identifier. Chunks are embedded using a biomedical dense encoder (MedCPT) to produce fixed-length vector representations. These vectors are indexed using FAISS to facilitate efficient nearest-neighbor searches throughout the corpus. A corpus index is created during this phase that can be reused and remains unchanged across all experiments.

3.3 Step 2: Dense Retrieval and Reranking

Encode a BioASQ question using the associated MedCPT query encoder, retrieve the 20 highest-scoring candidate passages for that question using dense retrieval through a FAISS index. Next, the 5 top-ranked candidate passages for a BioASQ question will be reranked and therefore will represent the candidate evidence set for the BioASQ question, supporting the most accurate precision scores possible for this evidence.

Notably, the configurations for retrieval i.e., embeddings, index, k/n values, and rerankers remain stable across all experimental conditions. Thus, any differences in downstream behavior can only be attributed to controlled variations in the evidence retrieved from the experiments, not to differences in the retrieval methods used.

3.4 Step 3: Controlled Evidence Perturbation (Stress Testing)

The main focus of this research is on the manipulated retrieval of evidence prior to generating an answer. Each question has three sets of evidence collected based on the top n retrieved passages and will undergo some modifications or manipulations:

i. Noise (Condition A)

Injecting irrelevant passages and/or near-miss passages (same disease, different drug/method) into the evidence set. The intensity of noise will vary as 30%, 50%, or 70% of original retrieved passages will be replaced.

ii. Conflict (Condition B)

Examples of supporting evidence and contradictory (e.g., passages that would appear to be contradictory) evidence (e.g., multiple instances where one set of supporting and/or opposing evidence has been published) will be included within the same set of evidence. Controlling for the ratio of supporting versus contradictory (e.g., 50/50 and 70/30) will provide a simulation of how contradictory (e.g., multiple instances) biomedical literature tends to be.

iii. Unanswerable Evidence (Condition C)

There are no answer-bearing passages left, so there will be no sufficient evidence to answer the question. There will be different settings based on the amount of removed evidence.

A baseline condition with clean, unmodified top-n passages is retained for comparison. Across all conditions, perturbations operate strictly at the passage level**,** and the retriever itself is not altered.

3.5 Step 4: Constrained Answer Generation

Use an LLM (large language model) that has been instruction-tuned for Answer Generation in the Pilot Study. At baseline, the generator that will be used is LlaMA3-8B-Instruct. However, the pipeline allows us to swap out other Open-Source Instruction LLMs and use the same Prompts and Evidence Constraints for generating responses through the Evidence-Based Rules of Prompting mentioned below:

- the LLM (Generator) must only use the information contained in the provided passage(s).
- for any Fact or Claim made, there must be at least one citation used to support it (e.g., the Identifier of the Citation); and
- If there is not enough Evidence to answer a question/claim, the LLM will explicitly state that it cannot make a guess.

Output format will follow the task definitions for BioASQ; for Yes/No Questions, it will return either a Yes or No label; for Factoid or List Questions, it will return either an Entity String or List of Entities; for Summary Questions, it will return a Paragraph summarizing the provided passage(s). Unless otherwise identified, experiments will hold the LLM generator fixed within each experimental run (to test the impact of perturbing the Evidence), and new experimental runs using the Evidence Retrieval and Generating model will be run when the LLM generator is changed.

3.6 Step 5: Multi-Level Evaluation and Error Analysis

Model Performance is evaluated through three distinct performance metrics:

i. Task Metrics

Evaluation Methods for Task-level metrics include exact match and F1 for fact-based and list-style questions; Accuracy and F1 for Yes/No questions; ROUGE-L for summary questions. All metrics must be consistent with BioASQ evaluation protocols.

ii. Reliability & Groundedness Metrics

Reliability & Groundedness metrics include Supported-Claim Rate, Citation Precision, and

Abstention Under Absence of Evidence metrics, as well as an NLI-based faithfulness metric that measures whether the provided evidence entails the generated claim.

iii. Retrieval and ranking metrics

The metrics we use in this process for independent evaluation of evidence retrieval are standard IR metrics (i.e., metrics used to measure the performance of information retrieval systems) computed against BioASQ gold documents and snippets. For this study, we will use MAP@k and MRR@k, which correspond to the average position of the correct evidence in the system's returned list and the average position of the first correct piece of evidence, respectively. In addition, whenever possible, we will also include nDCG@k to help account for the rank position of evidence in the retrieval list and support comparisons between retrieval methods. Using these three metrics allows us to assess retrieval failures separately from generation errors and to evaluate how retrieval/perturbation of evidence may impact ranking quality.

These 3 Performance Metrics allow for a nuanced diagnosis of RAG's various failure modes that aggregate scores alone cannot identify.

CHAPTER 4

PRIMARY SOURCES

This thesis uses official evaluation procedures, major biomedical literature repositories, and expert-curated biomedical question-answering datasets. These sources serve as the building blocks for the evaluation framework, retrieval index, and experimental corpus

4.1 BioASQ Benchmark Dataset

This research used the BioASQ datasets as its primary benchmark for biomedical question-answering data (Nentidis et al., 2025). Expert authors created biomedical questions (types: factoid, list, yes/no, and summary) in BioASQ, identified numerous peer-reviewed publications in PubMed with supporting snippets, and produced gold-standard responses for each question (Tsatsaronis et al., 2015).

Biomedical experts created the data; therefore, the BioASQ dataset benchmark is well established for testing biomedical QA. The way the data is formatted enables testing QA systems within rigorously defined experimental conditions.

All questions, answers, and supporting snippets are provided in English, and no translation is required; however, the domain knowledge used in the biomedical questions is expressed using the correct scientific terms. Therefore, the BioASQ data benchmark will serve as the 'ground truth' for:

i. Retrieval Evaluation
ii. Answer generation evaluation
iii. Analyzing the Faithfulness and Grounding of answers
iv. Conducting stress testing to examine the effects of slight variations in the retrieval stage on the results produced from the QA answer generation process.

4.2 PubMed Abstract Corpus (NCBI)

This thesis uses a retrieval corpus of PubMed abstracts (NCBI EFetch API). PubMed is the primary source of peer-reviewed biomedical literature and provides the evidentiary basis for retrieval-augmented generation (RAG) systems (Li et al., 2024).

Abstracts are accessed in XML format and converted into structured textual documents. These texts:

i. Will be chunked into passage-level segments
ii. Will be indexed and retrieved with dense/hybrid methodologies
iii. Will be used exclusively as evidence for generating answers.

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